

Blender -> Godot 3D Design Bible v1.0

World Scale

Blender Unit	1 m
Godot Unit	1 m
Export Scale	1.0
Model Grid	0.5 m
Large Modules	2 / 4 / 8 / 16 m

Human Ref

Height	1.80 m
Eye Height	1.68 m
Shoulder	0.48 m
Arm Span	1.80 m
Reach	2.30 m

Controller

Capsule H	1.80 m
Capsule R	0.30 m
Crouch H	1.20 m
Step H	0.25 m
Max Slope	45 deg
Stand Jump	1.30 m
Long Jump	3.5-4.0 m

Architecture

Interior Wall	3.0 m
Exterior Wall	3.5-4.0 m
Door	1.0 x 2.2 m
Double Door	2.0 x 2.2 m
Window Sill	0.9 m
Hall Min	2.0 m
Hall Comfy	3.0 m
Cover	0.9 m

Stairs / Ramps

Rise	0.18 m
Run	0.30 m
Step Limit	0.25 m
Collision	Ramp
Visual	Steps

Pivots

Wall	Bottom center
Floor	Center
Pillar	Bottom center
Door	Hinge
Window	Bottom center
Pipe	Center
Prop	Bottom center

Hierarchy

L0	Primitives
L1	Modules
L2	Assemblies
L3	Landmarks
Rule	Do not mutate lower-level modules for higher-level problems

Collision

Stairs	Ramp
Rubble	Simple hull
Pipes	Capsules/boxes
Rocks	Convex hull
Buildings	Primitive volumes

Scale Categories

0.1 m	Props
1 m	Furniture
10 m	Rooms
100 m	Buildings
1 km	Districts
10 km	Landmarks

Infrastructure

Pipe	Straight / Bend / Tee / Valve
Aqueduct	Pier + Arch x N + Cap
Bridge	Abutment + Span x N + Railings

Collections

Reference/ · Primitives/ · Architecture/ · Infrastructure/ · Nature/ · Props/ · Furniture/ · Lighting/ · Materials/ · Characters/ · Collision/

Materials

Concrete · Brick · Stone · Wood · Metal · Glass · Dirt · Grass · Sand · Water · Emissive

Naming

wall_2x3
wall_corner_inner
floor_4x4
pillar_round
arch_large
beam_4m
pipe_straight
pipe_bend_90
pipe_t
stair_18cm
lowercase, snake_case, metric when useful

Megastructures

Scale through composition, not giant cubes.
Door -> Room -> Building -> District -> Megastructure.
Large primitives: support columns, bridge spans, bulkheads, utility shafts, trusses, cable towers.

Philosophy

Gameplay metrics come first. · World scale is always metric. · Modules are reusable forever. · Build upward through composition. · Collision serves gameplay, not visuals. · Keep one Blender -> Godot pipeline. · Think in systems, not one-off assets.

Reference Scene

1.80 m mannequin · Character capsule · Door · Staircase · Hallway · Wall modules · Desk & chair · Floor grid · Lighting reference · Material spheres · Export test object